



USER MANUAL · VERSION 0.6

EZ2DEMOSCENE

Build seamless, stylised 3D loops the demoscene way: neon, mirrors, particles and retro pixels. No code needed.

For Windows, Linux, macOS, the web & Android · Covers the editor, the node graph, exporting and the command line

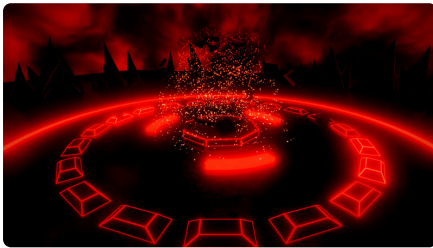
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1 Welcome

EZ2DEMOSCENE is a tool for making **short, perfectly looping 3D animations** in the style of the demoscene: glowing neon arenas, mirrored golden halls, glossy crystals orbited by debris, raymarched tunnels, kaleidoscopes and synthwave sunsets. The loops are for music visuals, VJ sets, wallpapers, social posts and GIFs.

You never write code or shaders. You pick a starting scene, then change it with sliders, colour pickers and menus. When you like it, you export an MP4, WebM, GIF or a folder of PNG frames.



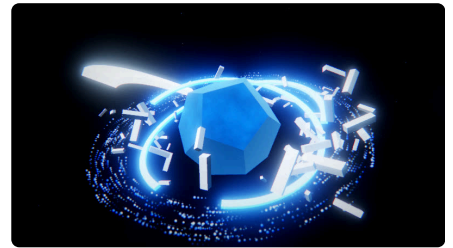
Neon Arena

Black gloss, red neon, crystal skyline.



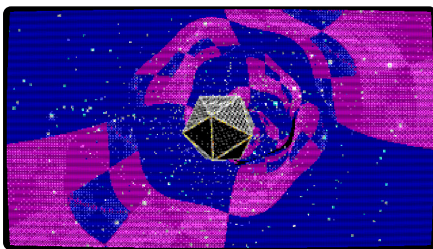
Gold Kaleido Room

Mirrored hall, orb walls, chandelier.



Orbiting Solid

Blue dodecahedron and a debris swarm.



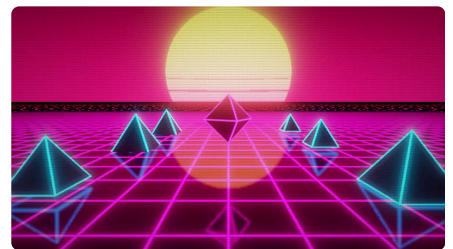
Retro Tunnel

XOR tunnel, EGA palette, CRT.



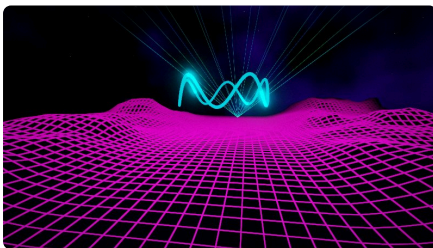
Plasma Kaleidoscope

Oldschool plasma, spinning tori.



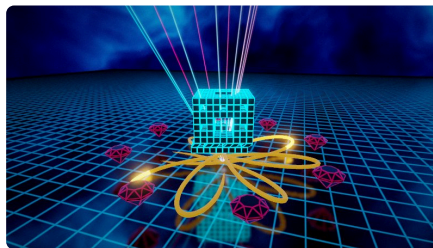
Synth Sunset

Striped sun, neon grid, pyramids.



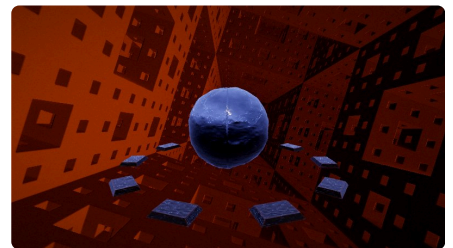
Vector Valley

Wireframe landscape, laser fans, neon ribbon.



Glitch Shrine

Glitching sponge, flashing gems, pulsing rose.



Sponge Dive

Menger sponge flight, breathing displaced orb.

The one big idea: everything loops

A scene is exactly **a whole number of beats** long, at a tempo you choose (BPM). Every motion in EZ2DEMOSCENE completes a whole number of times during that loop. That includes spins, orbits, pulses, particles, scrolling textures, drifting clouds and even film grain. So the last frame leads straight back into the first, with no jump.

Why this matters

You never have to "fix the loop" by hand. Whatever you change, the export repeats seamlessly. This is why many settings ask for a count of *turns per loop* or *cycles per loop* rather than a speed.

How a scene is built

A scene is a stack of **layers** drawn together:

- **Background:** the sky or environment behind everything (nebula, tunnel, plasma...).
- **Mirror floor:** a reflective ground plane, optionally with a glowing grid.
- **Shapes:** 3D objects, possibly copied many times in rings, grids, walls or swarms.
- **Particles:** sparks, stars, glitter, vortices.

On top of the layers sit the **camera**, the **light & fog**, and the **post effects** that give the final look (glow, kaleidoscope, retro palettes, CRT...).

2 Installing & launching

No install needed

You can also use EZ2DEMOSCENE in a web browser or on an Android phone. See chapter 15.

What you need

Item	Details
A graphics card	Any GPU with Vulkan, DirectX 12 or Metal support (most GPUs from 2015 onward). A software renderer also works, only slower.
Rust 1.95 or newer (<i>source builds only</i>)	Needed to build the app from source. Install it from <code>rustup.rs</code> , or update with <code>rustup update stable</code> .
ffmpeg (<i>optional</i>)	Needed only for MP4, WebM and GIF export. Download it from <code>ffmpeg.org</code> and make sure the <code>ffmpeg</code> command works in a terminal. PNG sequences need nothing extra.
Linux only	<code>sudo apt install libasound2-dev libxkbcommon-dev libwayland-dev</code>

The easy way: download a release

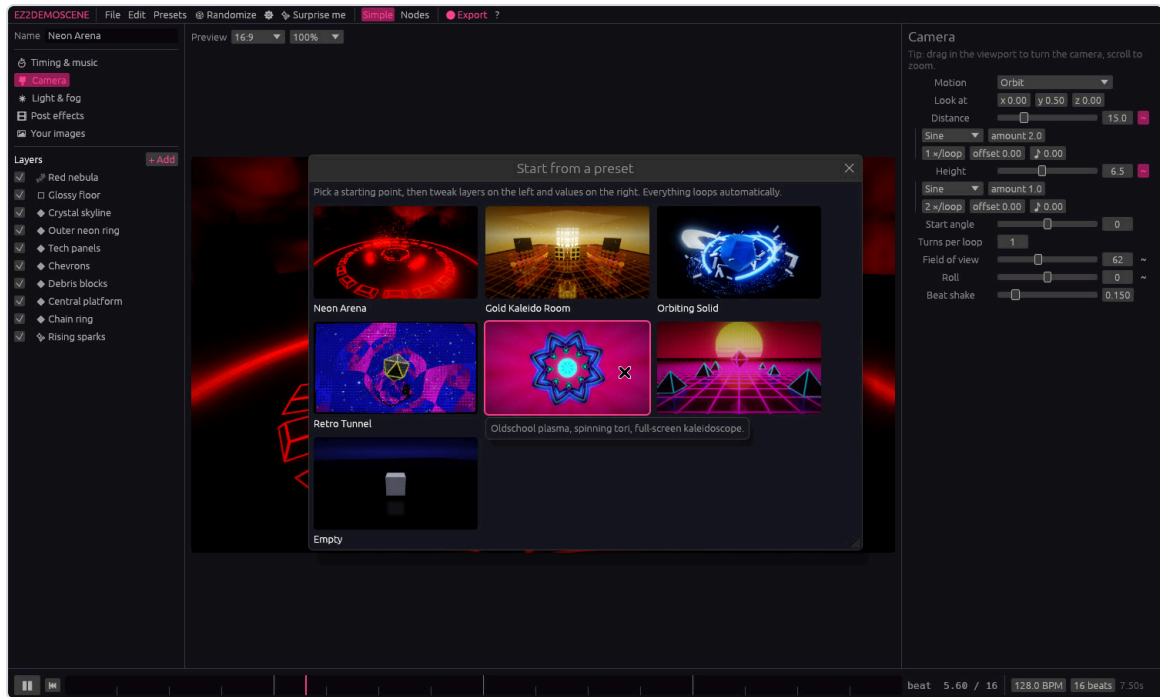
Ready-made downloads are published on the project's GitHub **Releases** page for Windows, Linux and macOS. Unzip the archive and run `ez2demoscene` (`ez2demoscene.exe` on Windows). The Windows and Linux downloads include **ffmpeg**, so video and GIF export work straight away. On macOS, install it with `brew install ffmpeg`. The archive also contains this manual, the presets and the texture pack.

Building from source

```
git clone https://github.com/Roganis/EZ2DEMOSCENE
cd EZ2DEMOSCENE
cargo run --release           # opens the editor
cargo run --release -- my.ez2.json  # opens a saved project
```

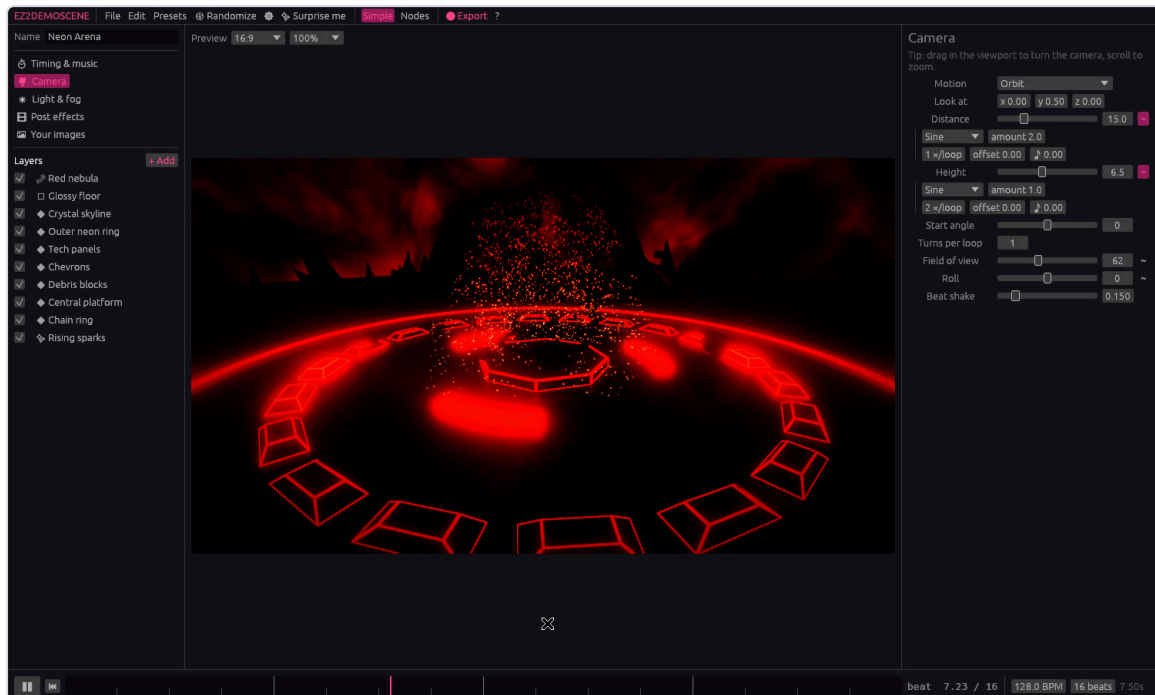
The first build downloads and compiles libraries, so it takes a few minutes. Later starts are instant. The finished program is `target/release/ez2demoscene` (`ez2demoscene.exe` on Windows), and you can copy it anywhere.

When the editor opens you are greeted by the **preset gallery**.



The preset gallery appears at start-up. Hover a card for a description, click to load it.

3 A tour of the window



The editor with the Neon Arena preset and the Camera selected.

Area	What it does
Top bar	File , Edit , Presets , 🎲 Randomize (with its ⚙ settings), Surprise me , the Simple / Nodes mode switch, ● Export and ? help.
Scene panel (left)	The project name ; the scene-wide sections (Timing & music , Camera , Light & fog , Post effects , Your images); and the layer list with + Add .
Viewport (centre)	A live preview. Drag to turn the camera and scroll to zoom. Above it you can pick the aspect ratio (16:9, 4:3, 1:1, 9:16, 21:9), the preview quality , a live fps readout, the Move / Rotate / Scale tools and the Grid switch (see 5.6). Click an object to select it. Status messages also appear here.
Inspector (right)	All the settings of whatever is selected on the left.
Timeline (bottom)	Play/pause, back to start, a scrubber with beat ticks (tall ticks mark every 4 beats), the current beat, BPM and loop length.

Tip

Hover almost any label in the inspector to see a short explanation.

4 Quick start: your first loop in five minutes

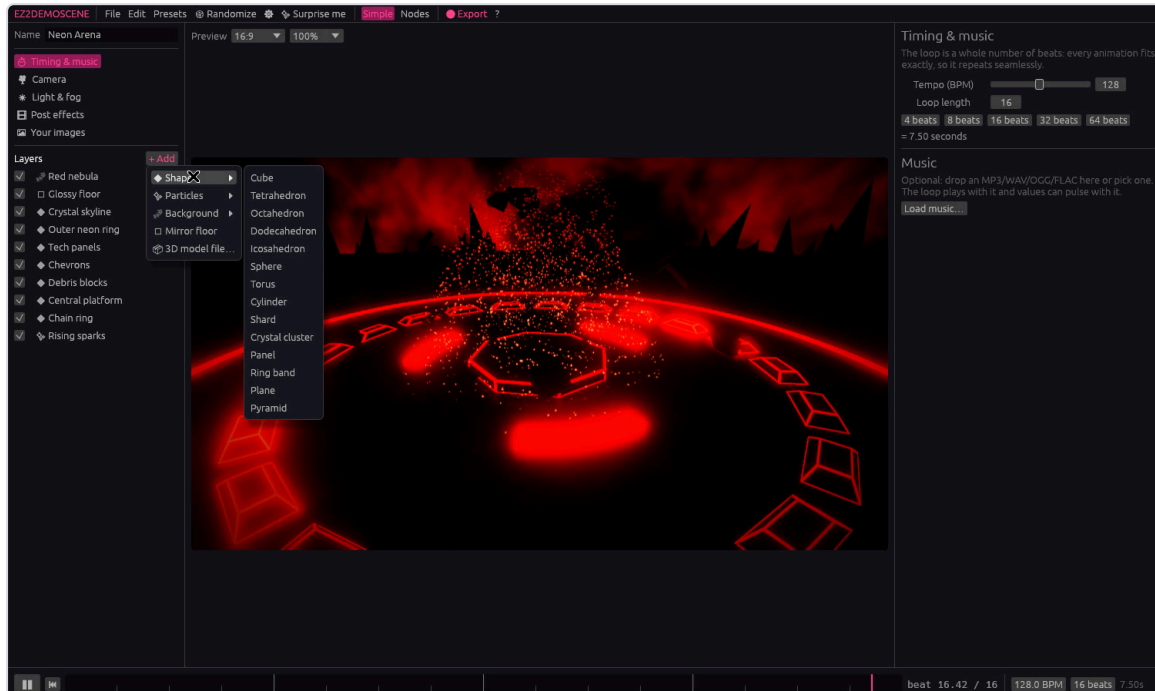
- 1 **Pick a preset.** In the gallery, click **Orbiting Solid**. The scene starts playing straight away.
- 2 **Change a colour.** Click the **Dodecahedron** layer on the left. In the inspector under **Material**, click the **Colour** swatch and pick a hot pink.
- 3 **Make it glow on the beat.** Still in **Material**, find **Glow** and click the ~ button next to it. Set the wave to **Pulse**, **amount** to 3 and **x/loop** to **8** (the loop is 8 beats long). It now flashes on every beat.
- 4 **Add particles.** Click + **Add** → ✨ **Particles** → **Vortex**. In the new layer, set **Colour (young)** to white and **Colour (old)** to pink.
- 5 **Add a retro touch.** Select **Post effects** on the left and tick **Retro palette**. Choose **PICO-8**.
- 6 **Export.** Press **Ctrl + E**. Pick **Animated GIF** (the size switches to 480×270, a good size for GIFs), choose a file with **Browse...** and press • **Export**.
- 7 **Save your project** with **Ctrl + S** to keep editing it later.

Made a mess?

Ctrl + Z undoes any change, including randomizing and loading presets.

5 Layers

The layer list works like the layers in a paint program. Each row has a **visibility** tick box and the layer's name. Click a row to edit it. Under the list, the buttons **↑ ↓ □ 🗑** move, duplicate or delete the selected layer, and **right-clicking** a layer shows the same options.



+ **Add** offers shapes, particles, backgrounds, a mirror floor and 3D model files.

5.1 Backgrounds

A background fills the whole screen behind the other layers and follows the camera. Every background has three colours (**Colour A** is usually the darkest), a **Motion / loop** count, a **Brightness** and a **Detail** (pattern scale).

Style	Description
Gradient sky	A soft two-colour sky with a horizon glow and a sun halo in the light direction.
Nebula clouds	Swirling fractal clouds with stars. The clouds drift on a closed path so they loop.
Starfield	Twinkling stars over faint coloured dust.
Raymarched tunnel	The classic demo tunnel flying forward through a winding tube. Pick a Wall texture (the XOR or checker textures are great).
Raymarched fractal	A volumetric "Kaliset" fractal space. It looks psychedelic and cosmic, and is heavy on slow GPUs.
Oldschool plasma	The 90s sine plasma.
Synthwave sun & grid	A striped sun sinking into a scrolling neon grid, with stars.

Raymarched sponge flight	Flying through an endless Menger sponge, a lattice of beams or a field of cubes. Heavy on phones: lower Quality if it stutters.
Raymarched ring corridor	Flying through glowing rings, squares or triangles that alternate between colours B and C.

Tip

You can stack two backgrounds, but the later one covers the earlier one. Usually one is enough.

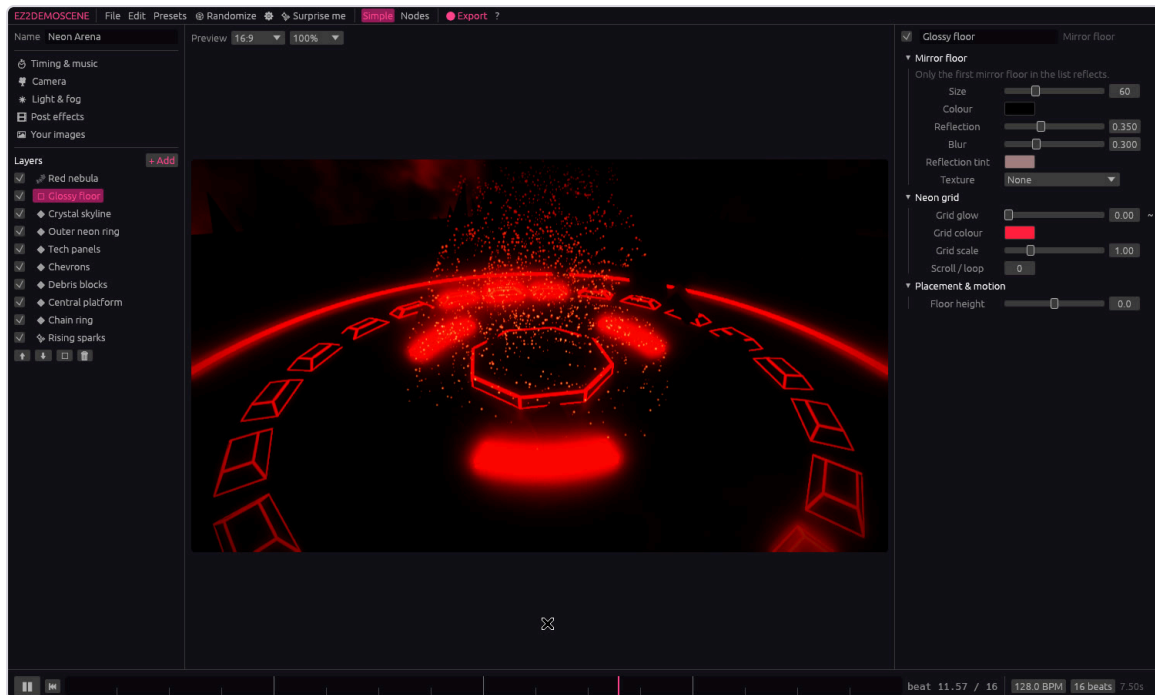
Raymarching settings

The four raymarched backgrounds (tunnel, fractal, sponge, rings) get an extra **Raymarching** section. Only the settings that matter for the chosen style are shown, and all of them except Variant, Wall pattern, Roll and Quality can be animated with ~.

Setting	Tunnel	Fractal	Sponge	Rings
Variant	Round, Square, Hexagon, Triangle, Flower	Kaliset, Crystal, Nebula	Menger sponge, Beam lattice, Cube field	Rings, Squares, Triangles
Size	Radius	Zoom	Cell size	Ring size
Twist	Twists the walls along the flight	n/a	Twists the space	Turns each ring a little more
Warp	Wall wobble	Fold (changes the whole look)	n/a	Thickness
Bend	How much the path winds	Drift	Path sway	Path sway
Glow	Light rings sliding past	Brightness	Edge glow	Glow
Fog ×	Depth fog (0 = none).			
Roll / loop	Whole turns of the view per loop.			
Quality	Raymarch steps (fractal: iterations). 0 uses the default; lower it for phones.			

The tunnel also has a **Wall pattern** (XOR, Checker, Rings, Stripes) used when there is no wall texture.

5.2 Mirror floor



Mirror floor settings.

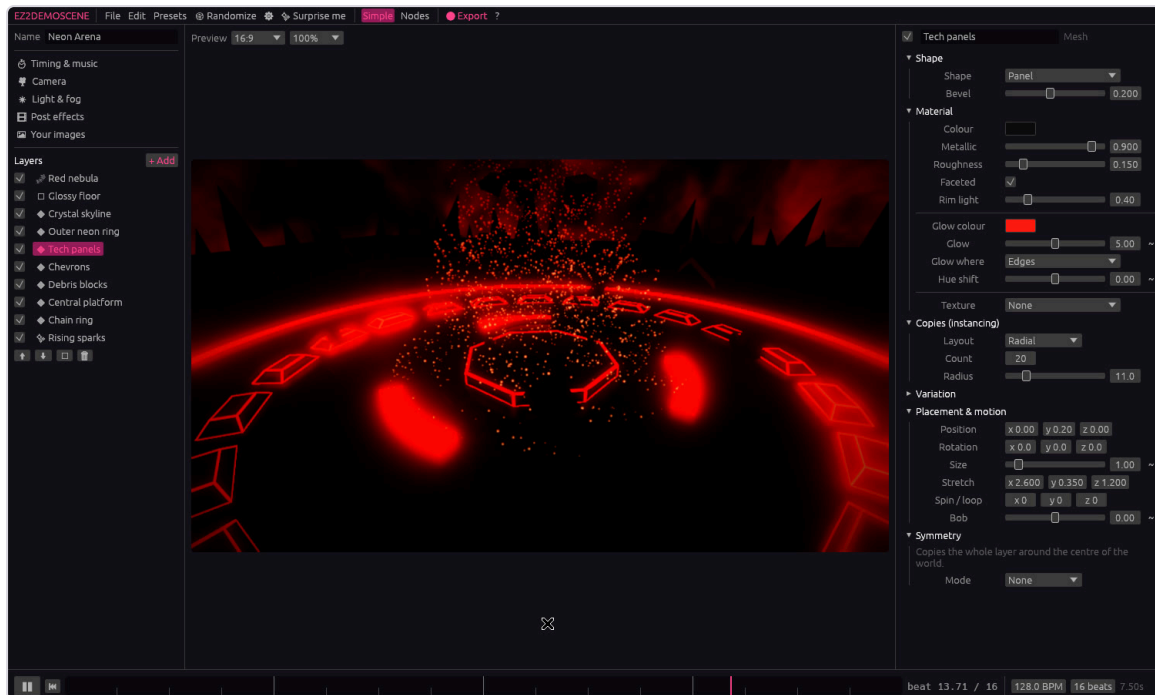
A large reflective ground plane. The scene is really reflected in it, including particles and the sky.

- **Reflection**: 0 is matte, 1 is a perfect mirror. **Blur** gives a frosted, polished-stone look. **Reflection tint** colours the reflection (gold for a gold room).
- **Texture** and **Tiling** put a pattern on the floor. The `led_grid` texture makes an LED dance floor.
- **Neon grid** draws glowing lines. Set **Grid glow**, **Grid colour**, **Grid scale** and **Scroll / loop** (cells per loop) for the synthwave "driving" effect.
- **Floor height** (in **Placement & motion**) moves the floor up or down.

One mirror

Only the **first** mirror floor in the list reflects. Extra ones are ignored.

5.3 Shapes



A shape layer (Tech panels): Shape, Material, Copies, Variation, Placement & motion, Symmetry.

In **Shape** choose one of 25 built-in shapes, or a 3D file:

Shape	Good for
Cube, Tetrahedron, Octahedron, Dodecahedron, Icosahedron	Classic demo solids. Try them with Faceted on and Glow where: Edges .
Sphere (Detail 0-5)	Orbs and bubbles. Detail 0 gives a faceted low-poly ball.
Torus	Rings and chain links (Thickness, Segments).
Cylinder (Sides)	Platforms and pillars. Eight sides makes an octagonal stage.
Shard, Crystal cluster	Sharp spikes and crystal mountains. Seed gives a different random shape.
Panel (Bevel)	Sci-fi plates, light panels, screens.
Ring band (Arc, Width, Height)	Neon rings and arcs. Set Arc below 360° for partial arcs and chevrons.
Plane, Pyramid, Cone	Simple building blocks.
Capsule (Length), Rounded cube (Roundness)	Soft, toy-like objects and pills.
Torus knot (Loops, Twists), Spring (Turns), Möbius strip	Twisty tubes and bands. Knots with more loops and twists get more intricate.
Star (Points, Inner radius), Gear (Teeth), Heart	Flat extruded logos and badges (Depth sets the thickness).

Gem (Facets)	A brilliant-cut diamond. Lovely with Glow where: Edges .
Menger sponge (Level 0-3)	The classic fractal cube. Level 3 has 8,000 cubes, so use it sparingly.
3D model file	Your own <code>.gltf</code> , <code>.glb</code> or <code>.obj</code> . It is centred and scaled to fit automatically, and uses the layer's material.

Copies (instancing)

One layer can show many copies of its shape. Pick a **Layout**:

Layout	Settings	Example
Single	n/a	One hero object.
Grid	Count x/y/z, Spacing	Speaker stacks, cube cities.
Radial	Count, Radius. Copies face outward.	Panels around an arena, crystal skyline.
Scatter	Count, Radius, Only surface, Seed	Floating debris, asteroid shells.
Orbit swarm	Count, Radius, Spread, Orbits / loop, Seed	Debris orbiting a planet.
Wall	Columns, Rows, Spacing, Curve	Orb walls, LED walls. Curve bends the wall into an arc.
Spiral	Count, Radius, Height, Turns	Chandeliers, DNA helices.

Variation (appears when there are copies) adds life:

- **Random tilt**, **Random size**, **Random hue** and **Random spin** make every copy a little different. Change **Seed** to reshuffle them.
- **Size wave** is a wave of growing and shrinking that travels across the copies.
- **Light chase** is a bright glow running along the copies, like marquee lights.
- **Waves / loop** sets how many times those waves pass per loop, and **Wavelengths** sets how many waves fit across all copies.

Placement & motion

Setting	Meaning
Position, Rotation	Where the layer sits and how it is turned (degrees).
Size	The size of the object, or of each copy . It does not change the distances between copies.
Stretch	Stretch per axis (make a cube into a slab or a pillar).
Spin / loop	Whole turns per loop around x, y and z. Negative numbers spin the other way.
Bob	Up/down offset. Animate it with <code>~</code> to make objects float.

5.4 Symmetry

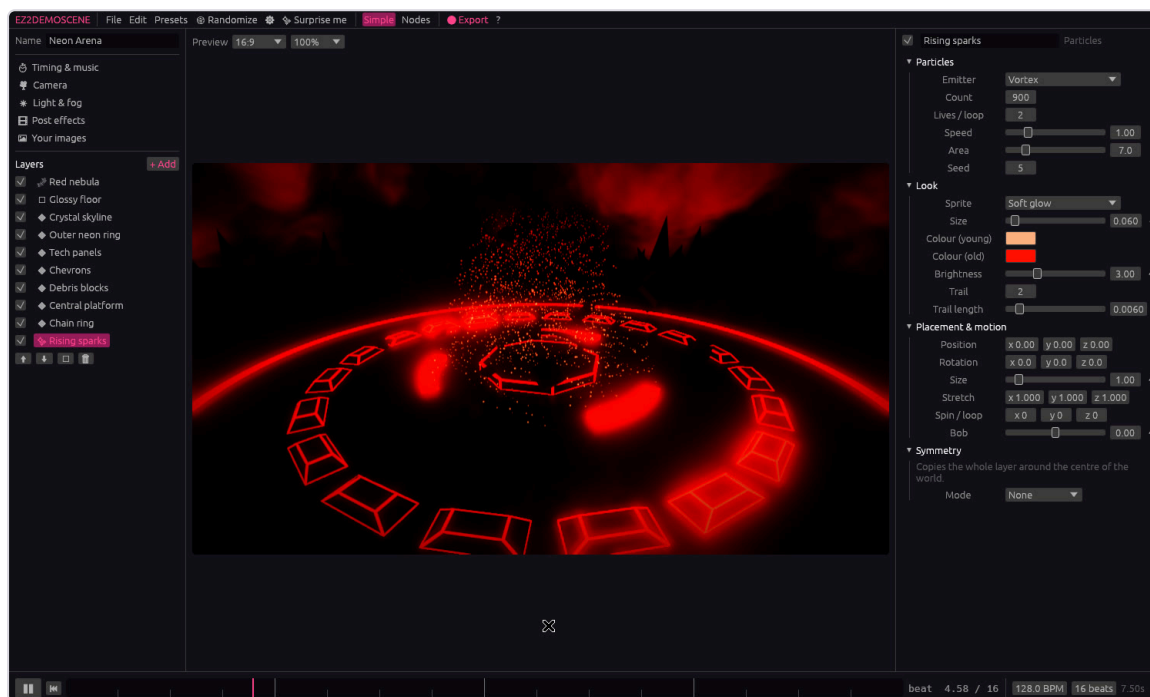
Symmetry copies the **whole layer** around the centre of the world:

- **Mirror X** (left/right), **Mirror Z** (front/back), **Mirror XZ** (four copies).
- **Radial**: N rotated copies around the vertical axis.
- **Kaleido**: N copies where every other one is mirrored, giving a kaleidoscope pattern.

Tip

Build one side of a room, then add **Mirror X**. The Gold Kaleido Room's orb walls, speakers and light panels are each built once and mirrored.

5.5 Particles



A particle layer.

Emitter	Motion
Burst	Explodes outward from the centre.
Sphere drift	Floats gently inside a sphere.
Ring orbit	Circles in a flat ring (Speed = whole orbits per loop).
Fountain	Shoots up and falls back.
Warp stars	Streams towards the viewer (hyperspace). Add a Trail for streaks.
Vortex	Spirals upwards like embers.
Snow / glitter	Falls slowly and sparkles.

Count is the number of particles. **Lives / loop** is how many times each particle is reborn in one loop, so higher means faster. **Area** is the size of the emitter. The look is controlled by **Sprite** (soft glow, pixel square, star, ring), **Size**, two colours (young → old), **Brightness** and **Trail** (ghost copies behind each particle).

Performance

Particles are cheap. Tens of thousands are fine on a normal GPU. Trails multiply the count, so $count \times (trail + 1)$ is what gets drawn.

5.6 Terrain

An endless landscape that rushes towards the camera, in the retro vector style. Add it with **+ Add** →  **Terrain**.

Setting	Meaning
Style	Wireframe (glowing grid lines only; things behind show through), Solid (lit ground) or Solid + lines .
Height, Hills, Roughness	How tall the mountains are, how many hills fit across the terrain, and how bumpy they are.
Valley	Flattens a road down the middle (0 = none), the classic synthwave look.
Scroll / loop	How many times the landscape scrolls past per loop. It repeats exactly, so the loop stays seamless. 0 stands still.
Line colour, Line glow, Ground colour	The look. Animate Line glow with a Pulse for beat flashes.
Texture, Tiles, Texture on lines, Chunky pixels	Maps a texture onto the ground (and optionally the grid lines). It moves with the landscape; Tiles is how many times it repeats across the terrain, a whole number so the loop stays seamless.
Size, Grid cells	Its extent and resolution. More cells is smoother but slower.

5.7 Laser beams

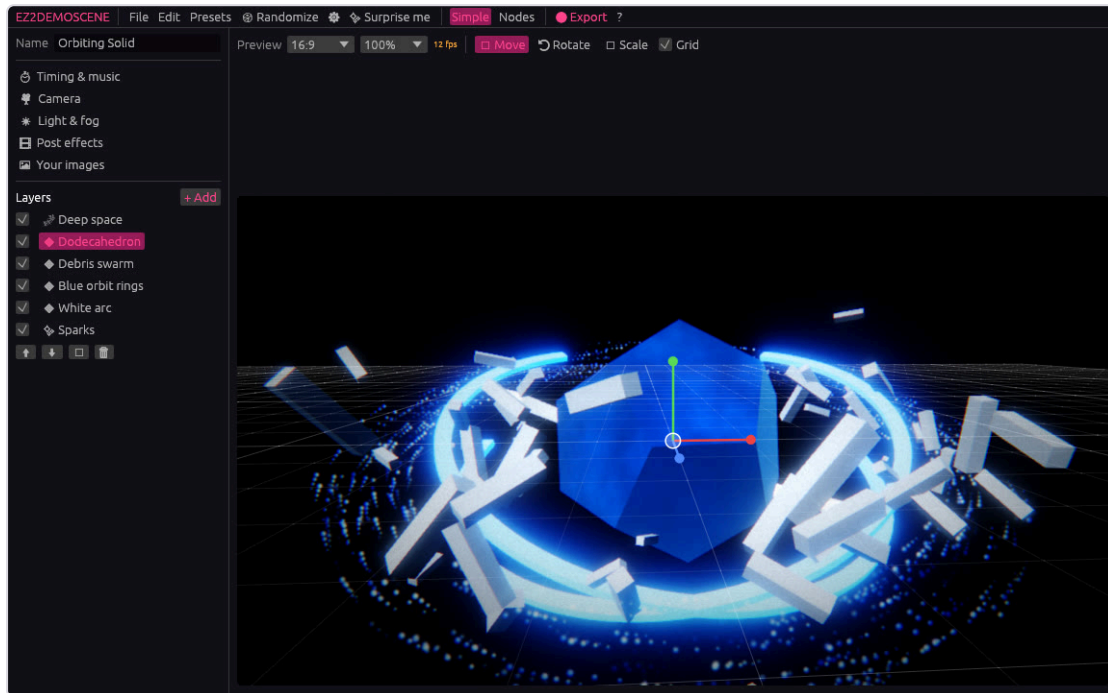
Club-style beams shooting from the layer's position along its up axis. Turn the layer with **Rotation** to aim them, and use **Mirror X** symmetry for a pair of laser rigs.

Setting	Meaning
Pattern	Fan (a flat fan that sweeps side to side), Rotating cone or Scatter (random directions that wobble).
Beams, Spread, Length, Width	How many beams, how wide the fan or cone opens (degrees), and their size.
Colour (first/last)	The beams blend from the first colour to the last.
Sweep, Sweeps / loop	How far the beams swing, and how often per loop (for the cone, also how fast it turns).
Beat strobe	Makes the beams flash on every beat and fade until the next one.

5.8 Neon ribbon

A glowing tube along a closed curve, with pulses of light running along it. Choose a **Curve** (**Lissajous**, **Knot**, **Figure eight**, **Wavy ring**, **Rose**) and its wave counts, then set **Pulses**, **Laps / loop** (negative runs backwards), **Pulse length** and **Pulse glow**. Ribbons reflect in mirror floors and can take symmetry.

5.9 Placing objects in the viewport



The Dodecahedron is selected: the Move handles appear on it, and the ground grid is on.

- **Select** a shape or particle layer by clicking it in the viewport, or in the layer list.
- Pick a tool above the viewport: **Move** (W), **Rotate** (E) or **Scale** (R).
- **Drag a handle.** Red is X, green is Y (up) and blue is Z. With Move, the white circle in the middle slides the object along the floor. With Rotate, drag one of the three rings. With Scale, the coloured squares stretch one axis and the middle circle changes the overall size.
- Hold **Ctrl** while dragging to **snap**: moves go in 0.25 steps, rotations in 15° steps and sizes in 0.1 steps.
- **Grid** (G) shows the ground grid, where one square is one unit.
- Dragging anywhere else still turns the camera, and the mouse wheel still zooms.

Tip

Pause first (Space) when placing spinning or orbiting things. Every handle drag counts as a single step for **Ctrl** + Z.

6 Materials, glow & textures

Surface

Setting	Effect
Colour	The base colour.
Metallic	0 looks like plastic or paint. 1 looks like metal, where reflections take on the colour (gold, chrome, copper).
Roughness	0 is a glossy mirror-like finish, 1 is matte. The glossy black look of Neon Arena is black + metallic 0.9 + roughness 0.15.
Faceted	Flat-shaded, low-poly look. Every face catches light separately.
Rim light	A glow along the silhouette that separates objects from dark backgrounds.

Glow (emission)

Glow colour and **Glow** (strength) make a surface emit light. Bright glow feeds the **bloom** post effect and produces the neon halo. **Glow where** decides which part glows:

Glow where	Result
Full	The whole surface glows (neon tubes, light panels).
Edges	Only the outlines of each face glow, for a Tron / wireframe-neon look. Works on the solids, panels, cylinders, shards and pyramids.
Stripes	Glowing horizontal stripes.
Texture	Bright parts of the texture glow (LED grids, circuits, tech-panel slits).

Hue shift rotates all the colours of the layer. Animate it with a Saw wave for rainbow cycling.

Relief: bump, normal and displacement maps

The **Relief** section of a shape layer makes its surface uneven. It uses the material's **Tiling** and **Scroll**, and the colour texture unless you pick a separate **Relief texture**.

Setting	Result
Mode: Bump	The texture's brightness is height: bright parts look raised, dark parts sunken (mortar between bricks, rivets, grid lines). Only the lighting changes, so it's cheap.
Mode: Normal map	The texture is a normal map (the purple-blue images from 3D tools and texture sites). Import one as an image and pick it as the relief texture.
Relief	Strength of the bump or normal map (animatable).
Displacement	Really moves the surface outwards by the texture brightness (negative pushes inwards). Animate it for pulsing, breathing shapes.

Subdivide

Adds geometry detail (each level splits every triangle in four), so displacement can show fine patterns.

Tip

Displacement looks best on smooth shapes (sphere, torus, capsule, rounded cube). Faceted shapes such as the cube or the gem open up at their edges, because each face moves along its own direction.

Glitch

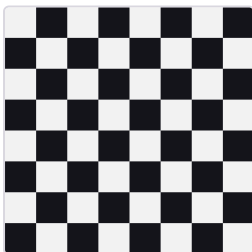
The **Glitch** section of a shape layer corrupts its geometry, like a broken signal. Set an **Amount** and a **Style**:

Style	Result
Jitter	The corners shake to random positions; the shape crumples without tearing.
Slices (VHS)	Horizontal bands of the object jump sideways.
Shatter	The faces fly apart along their directions.

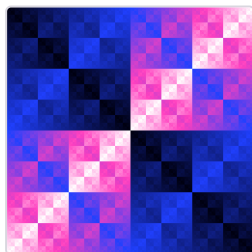
Changes / loop is how many new random patterns appear per loop, and **Chance** is how many of those moments actually glitch (1 = all the time, 0.3 = short bursts). Like everything else, the glitch pattern repeats exactly every loop. Animate **Amount** with a Pulse for hits on the beat.

Textures

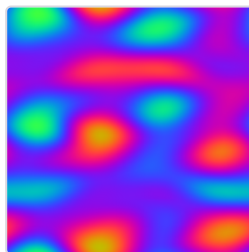
Pick a **Texture** from the built-in retro pack or from your own images. **Tiling** repeats it, **Scroll / loop** slides it by whole tiles per loop, and **Chunky pixels** keeps the pixels sharp instead of smoothing them.



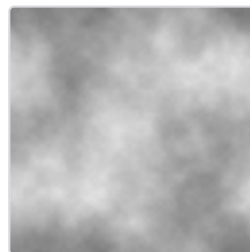
checker



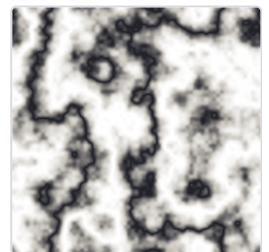
xor



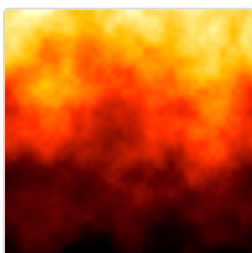
plasma



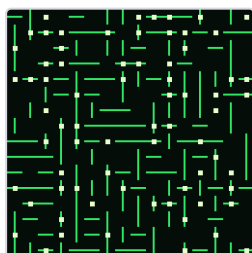
noise



marble



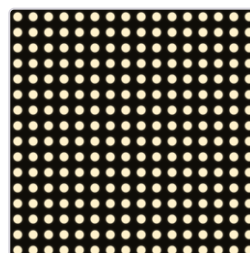
fire



circuit



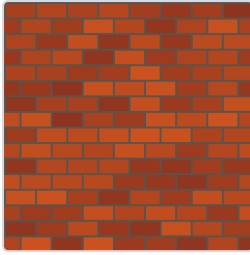
tech_panel



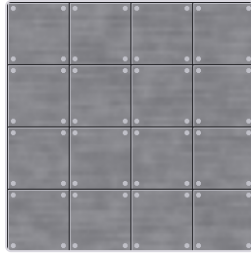
led_grid



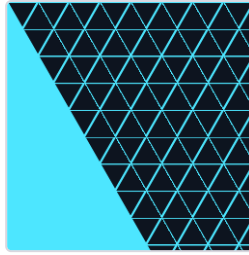
speaker



brick



metal_plate



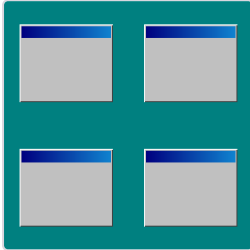
hex



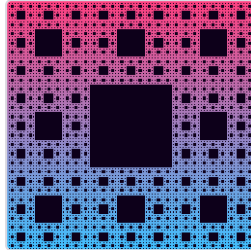
stripes



copper



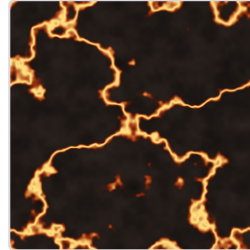
win9x



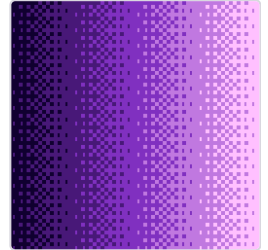
sierpinski



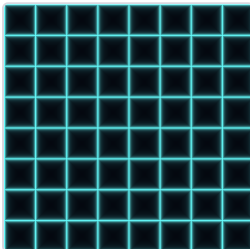
stars



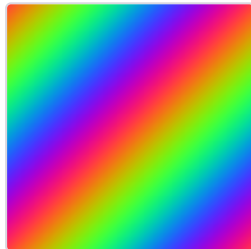
lava



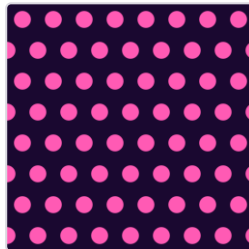
dither



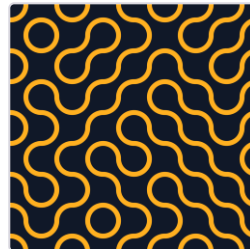
tron_grid



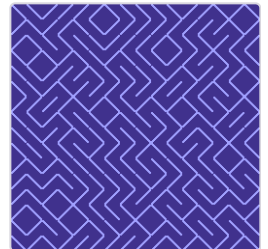
rainbow



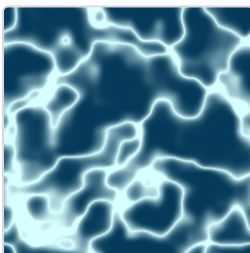
polka



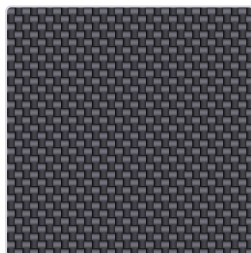
truchet



maze



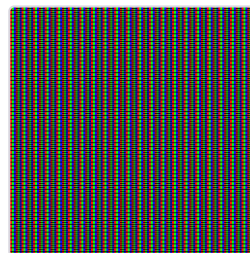
caustics



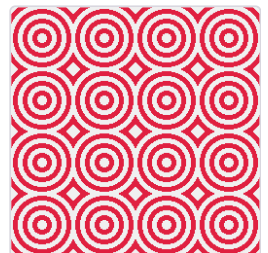
carbon



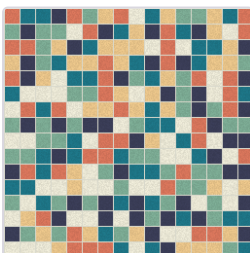
wood



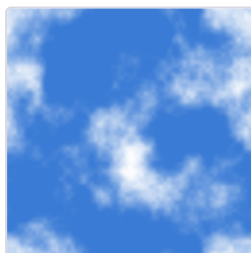
crt_mask



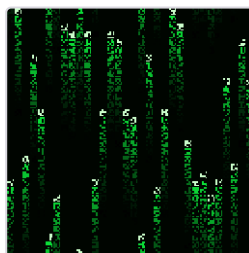
rings



mosaic



clouds



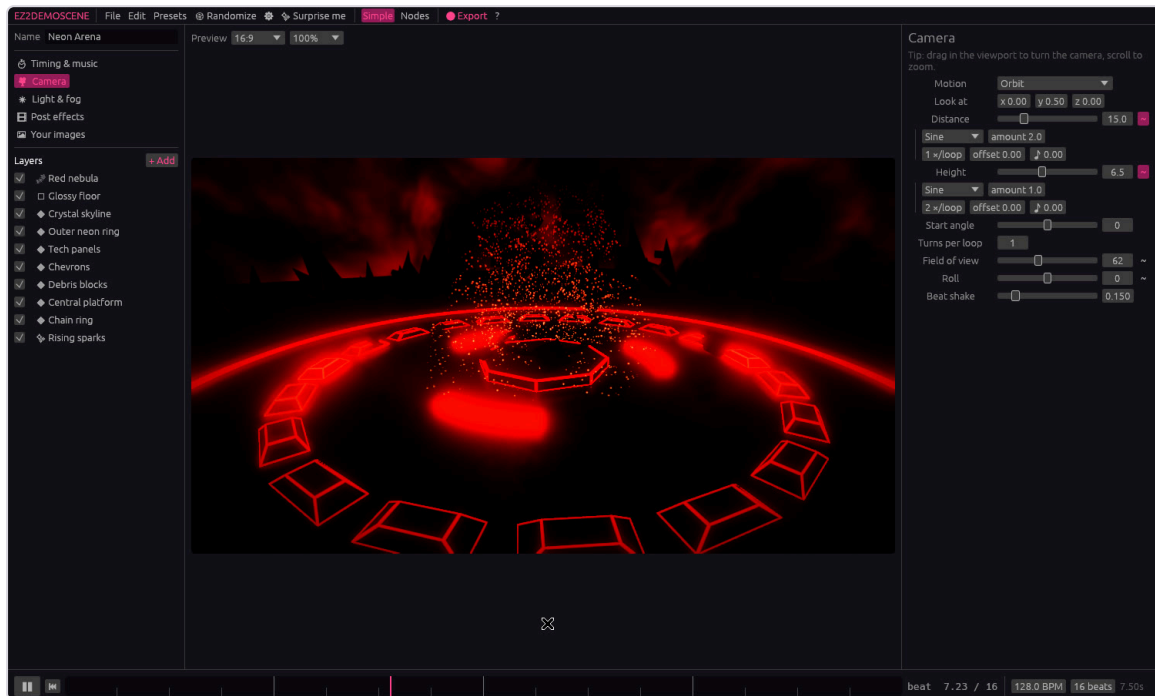
matrix

Your own images & "retro-ize"

Import images (PNG, JPG, BMP, GIF, TGA) in any of these ways: from a texture menu (**Import image...**), from **File → Import image...**, or by dragging them onto the window. A dropped image is applied to the selected shape. All your images are listed under **Your images**, where you can tick **Retro-ize**. This shrinks the image to **Max size** pixels and remaps it to a classic **Palette** with **Dithering**, turning any photo into a 90s texture.

7 Animating anything

Every value with a ~ button next to it can move over time. The button turns pink when the value is animated. Click it to open the animation panel:



In the Camera section, Distance and Height are animated: their panels show wave, amount, x/loop, offset and ~.

Control	Meaning
Wave	The shape of the movement, in three groups (see below). Hover a shape in the menu for a description.
amount	How far the value moves from the slider value. 0 means not animated. Picking a shape or a rhythm on a still value gives it a starting amount so you see something straight away.
~	Rhythm menu: twice per beat, every beat, every 2 beats, every bar, every 2 or 4 bars or once per loop . It sets x/loop for you from the loop length.
x/loop	Whole cycles per loop. The number of beats in the loop hits every beat . Negative numbers run backwards.
offset	Shifts the timing (0-1 of a cycle), to stagger several animations.
♪	Adds the music's loudness (see chapter 10).

Below the controls, a small **graph** shows the value over the whole loop, with beat ticks and a dot at the current moment.

Group	Shapes	Use them for
-------	--------	--------------

LFO	Sine, Triangle, Saw up, Saw down, Square	Smooth or hard back-and-forth movement around the slider value.
Beat fades	Pulse (hit), Exp fade in, Exp fade out, Linear fade in, Linear fade out, Swell in & out	Envelopes that run from 0 to the full amount (or back) once per cycle. Set J → every beat for hits that decay (Exp fade out), build-ups that drop on the beat (Exp fade in), or breathing swells.
Random	Sample & hold, Smooth random, Drunken walk	A new random value each cycle (held, or gliding smoothly), or a random walk that wanders and always finds its way back by the end of the loop.

Almost every number can be animated this way, including **post effects** (bloom threshold, pixel size, dither, CRT scanlines and curvature, contrast, saturation, vignette, grain, beat flash, kaleidoscope angle), materials (metallic, roughness, rim light, tiling), light, camera swing and shake, particle speed and area, mirror reflectivity and blur, laser spread, length, width and sweep, terrain roughness and valley, and ribbon pulses.

Recipes

- **Pulse on every beat:** Glow → Pulse, amount 3, $\times/\text{loop} = \text{loop beats}$ (e.g. 16).
- **Breathing camera:** Camera Distance → Sine, amount 2, $\times/\text{loop} 1$.
- **Kick zoom:** Kaleidoscope Zoom → Pulse, amount 0.2, $\times/\text{loop} 16$.
- **Rainbow:** Hue shift → Saw, amount 0.5, $\times/\text{loop} 1$.
- **Floating object:** Bob → Sine, amount 0.5, $\times/\text{loop} 2$.
- **Strobe:** Glow → Square, $\times/\text{loop} 32$.
- **Beat hit:** Glow → Exp fade out, J every beat.
- **Build-up:** Bloom threshold → Exp fade in, J every 4 bars (negative amount lowers it towards the drop).
- **Nervous camera:** Camera Height → Drunken walk, J every beat, amount 1.

Shake

Placement & motion → **Shake** jolts a layer in a random direction. **Shake distance** and **Shake turn** set how far, and **New direction** how many times per loop a new random direction is picked. Give the distance a fade for a hit that settles: the **Beat jolt** button does exactly that (Exp fade out, every beat). The **Jitter** node can drive the same shake for everything connected to it (chapter 12).

Blink / strobe

Every layer has a **Blink / strobe** section that makes the **whole layer** appear and disappear, or flash:

Mode	What it does
Blink	On and off in a regular rhythm. On time is the fraction of each blink it stays visible.
Random blink	On and off at random moments. Chance on sets how often it is visible.

Flash

Stays visible but its glow flashes and fades. **Dim level** is the brightness between flashes.

Times / loop sets the rhythm (16 = every beat of a 16-beat loop) and **Offset** shifts it, so two layers can blink in turn.

8 Camera, light & fog

Camera

Setting	Meaning
Motion	Orbit circles the target Turns per loop times. Pendulum swings back and forth by Swing degrees. Static stays put, though its values can still be animated.
Look at	The point the camera aims at.
Distance, Height	How far away and how high the camera is. Both are animatable.
Start angle	Which side the camera starts from.
Field of view	Zoom lens: 20-35 for a telephoto look, 60-90 for wide and dramatic.
Roll	Tilts the horizon. Animate it for a dizzy rave feel.
Beat shake	A small camera kick on every beat.

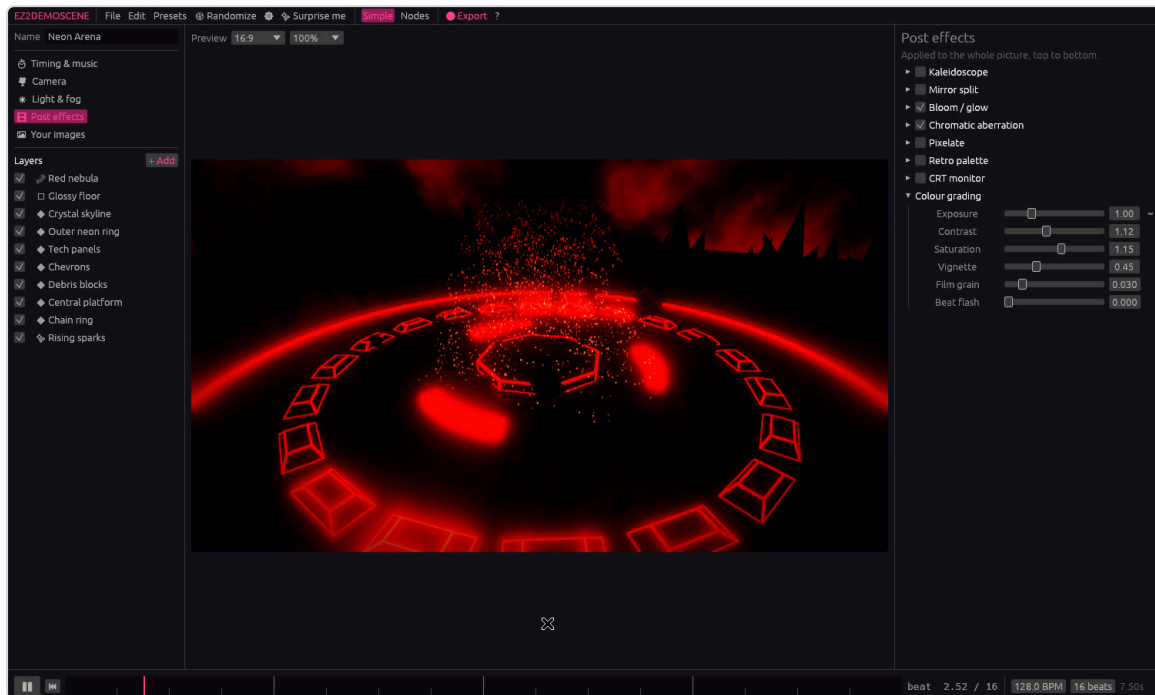
Mouse

Dragging in the viewport changes **Start angle** and **Height**. Scrolling changes **Distance**.

Light & fog

- **Fog colour** / **Fog density**: distant things fade into this colour. The fog colour is also the "empty space" colour, so match it to your background.
- **Sky light** / **Ground light** / **Ambient**: soft light from above and below. The sky colour is also what glossy surfaces reflect, which gives black gloss its highlights.
- **Sun colour**, **Sun strength**, **Sun direction**: the main light.

9 Post effects



Post effects. Tick a section's box to enable it, and click its arrow to open its settings.

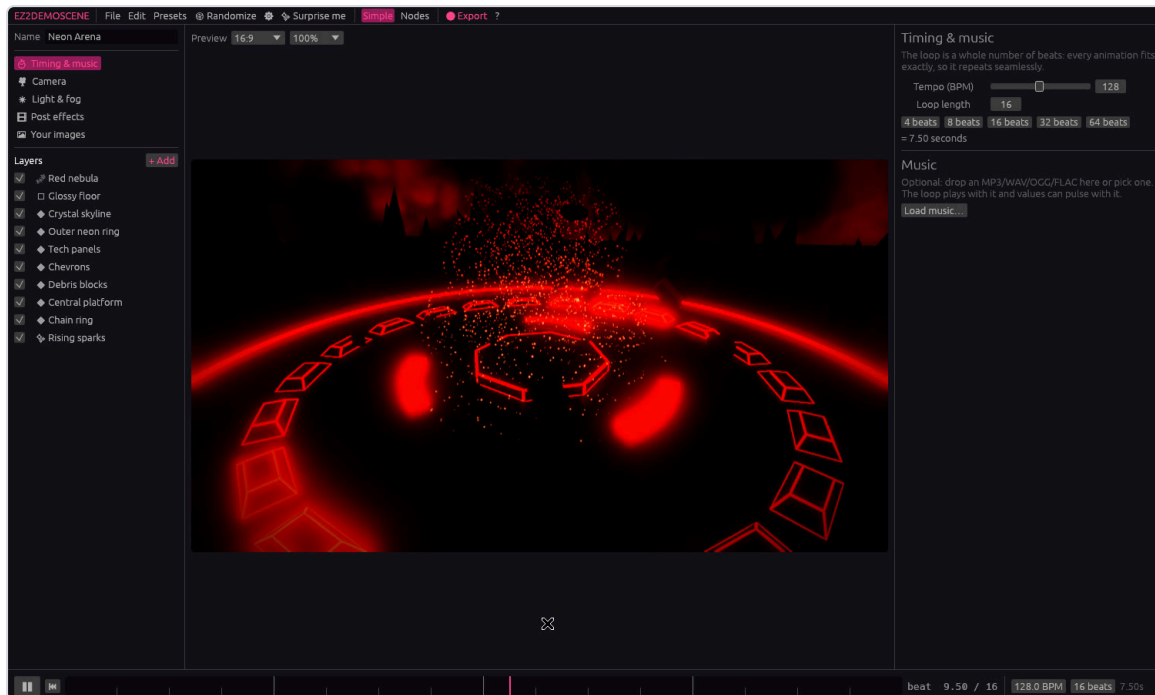
Effect	What it does
Kaleidoscope	Folds the whole picture into mirrored wedges. You can set Segments, Angle, Turns per loop, Zoom (animatable) and Centre.
Mirror split	Mirrors one half of the picture onto the other (left → right, top → bottom, or quad). This is instant symmetry.
Bloom / glow	Bright things spill light. Intensity sets the strength, Threshold how bright something must be to glow, and Spread how wide the halo is.
Chromatic aberration	An RGB colour split towards the edges, like a cheap lens. Pulse it on the beat.
Pixelate	Fat pixels (the size is measured at 1080p, so previews match exports).
Retro palette	Reduces the image to a classic computer palette with ordered (Bayer) dithering.
CRT monitor	Scanlines, screen curvature and a VHS wobble.
Colour grading	Exposure, Contrast, Saturation, Vignette (dark corners), Film grain and Beat flash (a white flash on every beat).

Palettes

Palette	Colours
EGA (16)	

CGA (4)	
Commodore 64 (16)	
Game Boy (4)	
PICO-8 (16)	
Amiga copper (16)	
ZX Spectrum (8)	
VGA cube (216)	Every colour quantised to 6 levels per channel, a subtle retro banding.
Green phosphor (4)	

10 Timing & music



Timing & music.

Tempo and loop length

Tempo (BPM) and **Loop length** (in beats) set how long the loop is. For example, 128 BPM × 16 beats = 7.5 seconds. The quick buttons set 4, 8, 16, 32 or 64 beats. These values also appear in the timeline.

Because every animation is counted per loop, **changing the BPM speeds the whole scene up or slows it down**, and it stays in sync.

Adding music

Use **Load music...** or drop an MP3, WAV, OGG, FLAC, M4A or AAC file onto the window. Then:

- The music plays with the loop and restarts at the loop point. Scrubbing the timeline seeks the music.
- Its loudness is drawn in the timeline, and brighter bars mean more bass.
- Any animated value can follow the music with the 🎵 amount in its ~ panel.
- **Detect tempo from length** sets the BPM so the whole track fits the loop length. This is useful for music written as a loop.
- MP4 and WebM exports include the music (you can turn this off in the export dialog).

Tip for musicians

Set the BPM to your track's tempo and the loop length to its bar count × 4. The visuals then land exactly on your beats.

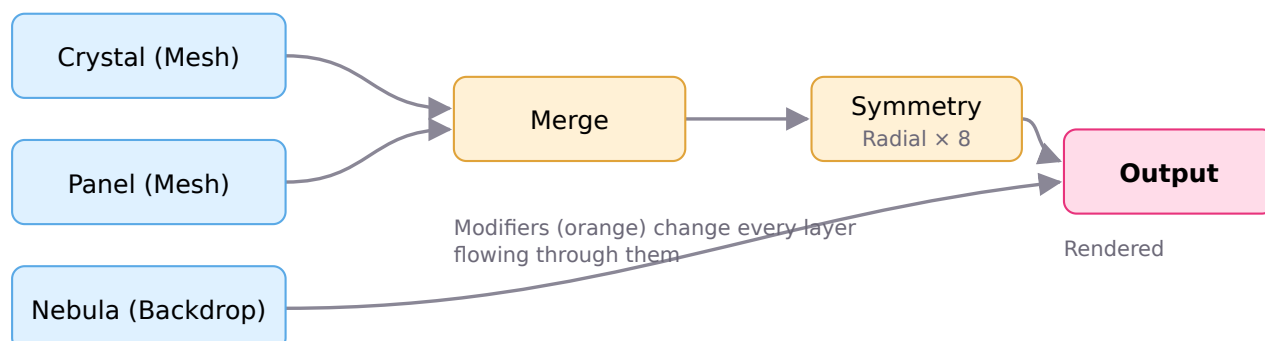
11 Randomize, Surprise me & undo

- 🎲 **Randomize** shuffles the current scene: one harmonious hue rotation of all colours, new shapes and layouts, new motion and a few post-effect changes. Every click gives a new variation.
- ⚙️ opens the **Randomizer** settings. Choose what may change (Colours, Shapes & layouts, Motion, Post effects), how strongly, and the **seed**. The same seed always gives the same result.
- **Surprise me** picks a random preset and randomizes it strongly.
- **Undo / redo** (`Ctrl + Z` / `Ctrl + Shift + Z`) covers everything, so randomize freely.

12 Nodes mode (advanced)

Click **Nodes** in the top bar to build the scene as a **graph**. Layers become **source nodes** that flow through **modifiers** into an **Output**. Everything connected to Output is rendered.

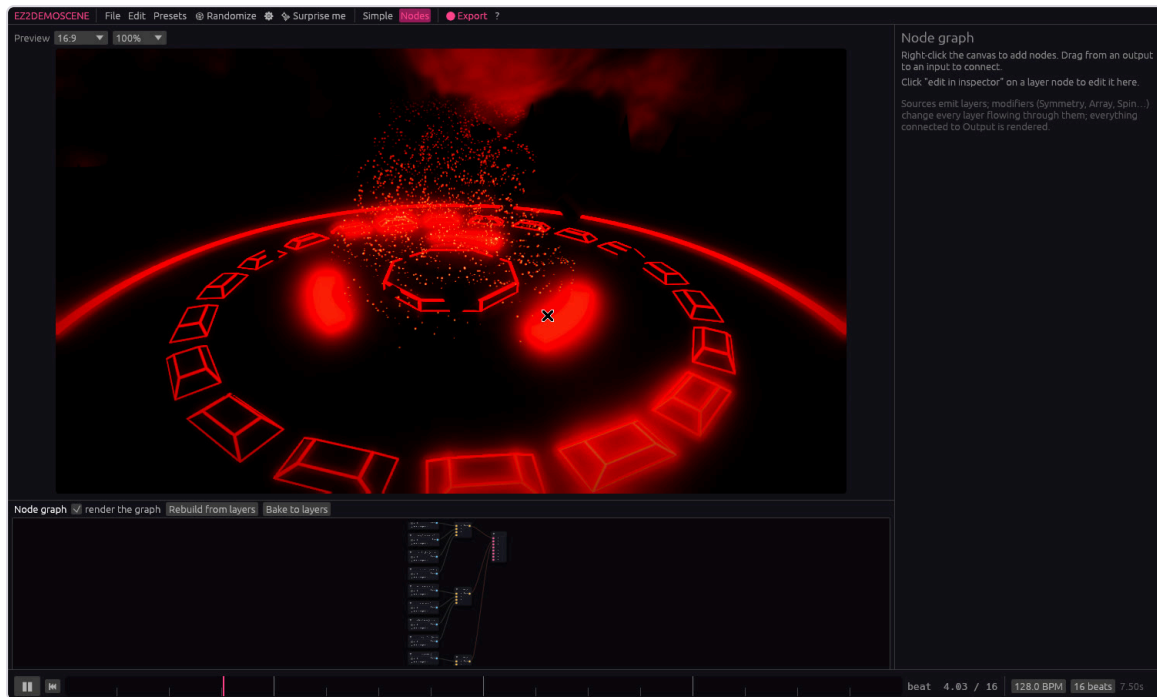
Sources (blue) emit layers



The crystal and panel layers are merged, turned into 8 radial copies, and sent to Output together with the nebula.

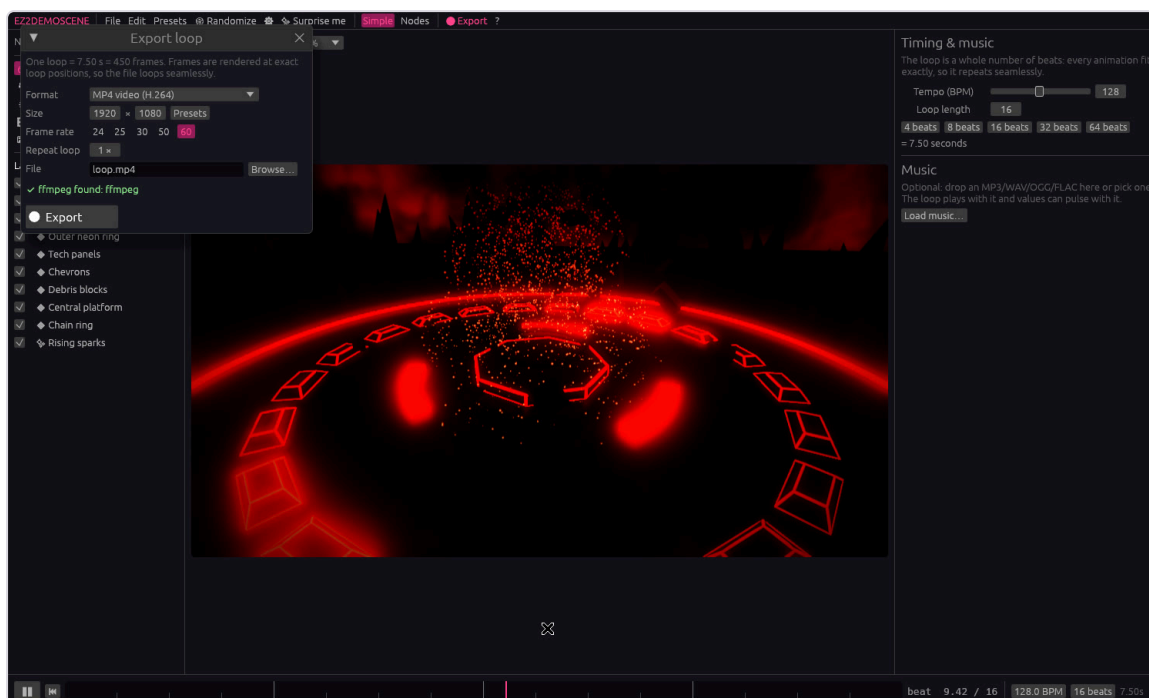
Node	What it does
Source	One layer. Click edit in inspector to edit it on the right with the usual controls.
Symmetry	Replaces the symmetry of every incoming layer.
Offset	Moves every incoming layer.
Spin	Adds turns per loop.
Scale	Multiplies the size.
Tint	Rotates hues and scales the glow.
Array	Repeats the incoming layers N times, stepping the position and rotation for each copy.
Jitter	Moves and turns every incoming layer by a random amount (seed reshuffles). With re-roll at 0 it scatters the layers once; above 0 it becomes a beat-synced shake that picks a new direction that many times per loop. Move and Turn have ~ , so fades work here too.
Mirror	Adds a mirrored copy of every incoming layer, across the X, Y or Z plane at a chosen position.
Strobe	Sets the Blink / strobe of every incoming layer (chapter 7).
Colour / material	Sets the glow colour, multiplies the glow, and for shapes can set a texture, a neon wireframe look and a glitch.
Merge	Combines up to 4 streams.
Output	Up to 8 inputs. Everything here is drawn.

Working with the graph: right-click empty space to add nodes, drag from an output dot to an input dot to connect, and right-click a node to **Duplicate** or **Delete** it. The toolbar has three controls: **render the graph** switches between the graph and the plain layer list; **Rebuild from layers** starts a new graph from your layers; and **Bake to layers** turns the graph's result into ordinary layers and returns to Simple mode.



Nodes mode: the viewport on top, the graph below, the inspector on the right.

13 Exporting your loop



The export dialog (`Ctrl + E`).

Option	Notes
Format	MP4 (H.264) works everywhere. WebM (VP9) is for the web. Animated GIF is for chats and forums. PNG sequence writes one PNG per frame into a folder and needs no ffmpeg; it is ideal for video editors and VJ software.
Size	Type a size or use Presets : 720p, 1080p, 1440p, 4K, Square, Vertical (phone), GIF small, Retro 4:3.
Frame rate	24, 25, 30, 50 or 60 fps. The dialog shows how many frames one loop needs.
Repeat loop	Makes the file longer by repeating the loop. Useful for platforms that don't loop videos.
Music	Include the music track (MP4/WebM only).
File / Folder	Where to save. Use Browse....

Press **Export**. A progress bar shows the frames being rendered, and you can keep working or press **Cancel**. Export renders every frame at full quality (anti-aliasing and full-resolution effects) even if your preview is set lower.

GIF tips

GIFs get big quickly. Use 480×270 at 25–30 fps, keep loops short (8 beats), and avoid film grain, which makes GIFs much larger.

"ffmpeg not found"?

Install ffmpeg, then either make sure `ffmpeg` runs in a terminal or paste the path to the `ffmpeg` program into the box in the export dialog and press **Check**.

Stills

File → Save still image... saves the current frame as a PNG at the export size.

Command line

You can render without opening the window, which is handy for batch jobs:

```
ez2demoscene --list-presets
ez2demoscene --export friend_scene.ez2pack loop.mp4    # packs work too
ez2demoscene --render "Neon Arena" still.png --phase 0.25 --size 1920x1080
ez2demoscene --export "Gold Kaleido Room" loop.mp4 --size 1920x1080 --fps 60 --repeats 4
ez2demoscene --export my.ez2.json frames/              # PNG sequence into a folder
ez2demoscene --write-presets my_presets/               # presets as editable project files
ez2demoscene --write-textures my_textures/             # the retro texture pack as PNGs
```

`--phase` is a position in the loop from 0 to 1 (0.5 is the middle).

14 Saving, sharing & your library

Project files

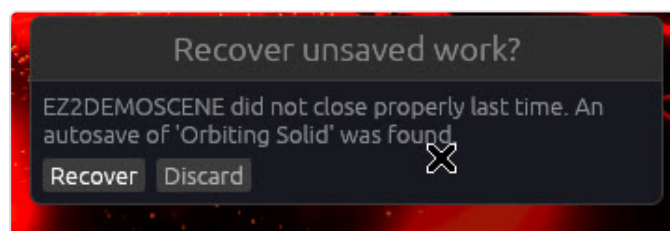
File → Save (**Ctrl** + **S**) writes a `.ez2.json` project. Models, images and music stored **inside the project's folder** (or its sub-folders) are remembered **relative to the project**. You can move or copy the whole folder, even to another computer, and everything is still found. Files elsewhere on your disk are remembered by their full path.

Packs: one file with everything

File → Save as pack (.ez2pack)... writes a single file containing the project **and copies of every model, image and music file it uses**. It's the best way to send a scene to a friend or archive it. Open a pack with **File → Open...** or by dropping it on the window. It is unpacked into your library folder, and **Save** then asks where to keep the project. If some files were missing when packing, the status bar lists them.

Autosave & crash recovery

While you work, unsaved changes are autosaved every 30 seconds. If the program or computer crashes, the next start offers to **recover** them:

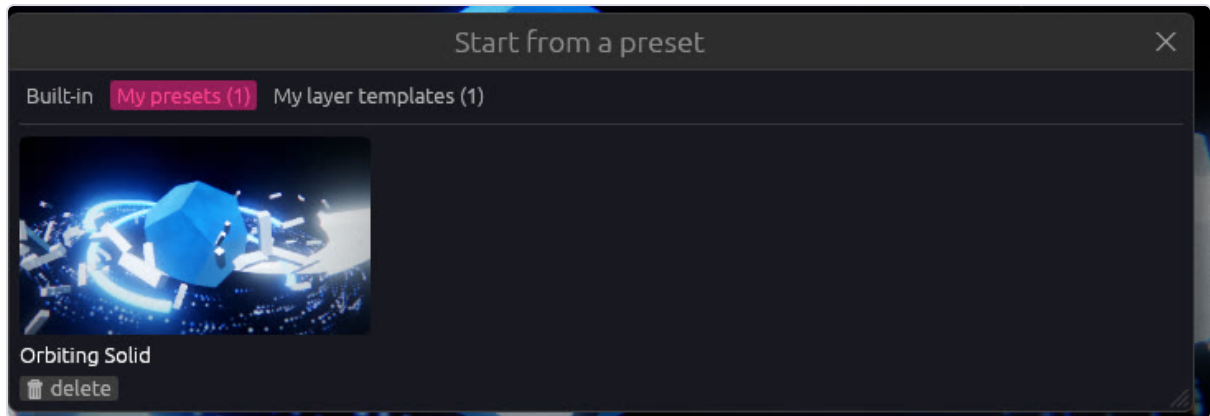


Shown after a crash. **Recover** brings back the scene; **Discard** throws the autosave away.

After recovering, save the project somewhere with **Ctrl** + **S**. When you quit normally, the autosave is deleted.

My presets

File → Save as my preset... adds the current scene to the gallery with a thumbnail of the current frame. Find it under **Presets → My presets**. Clicking one opens it as a new untitled project, so your preset is never overwritten by accident. The  **delete** button removes it.



The gallery's **My presets** tab.

Layer templates

Built a great particle effect or neon ring? **Right-click the layer → Save as template**. It then appears in **+ Add → ★ My templates** in every project, and also in the node graph's right-click menu. Manage your templates in **Presets → My layer templates**.

Where is my library?

Your presets, templates, unpacked packs and autosave live in your user data folder:

`%APPDATA%\ez2demoscene\EZ2DEMOSCENE\data` on Windows, `~/.local/share/ez2demoscene` on Linux, and `~/Library/Application Support/org.ez2demoscene.EZ2DEMOSCENE` on macOS. Copy it to back everything up.

15 Phones, tablets & the web

EZ2DEMOSCENE also runs **in a web browser**, with nothing to install. It's the same editor with the same presets, layers, node graph and exports. Open `https://roganis.github.io/ez2demoscene/` in Chrome, Edge or Firefox on a computer, or in Chrome on an Android phone or tablet. On Android you can also install the **EZ2DEMOSCENE app (APK)**, which bundles the same editor and works offline.

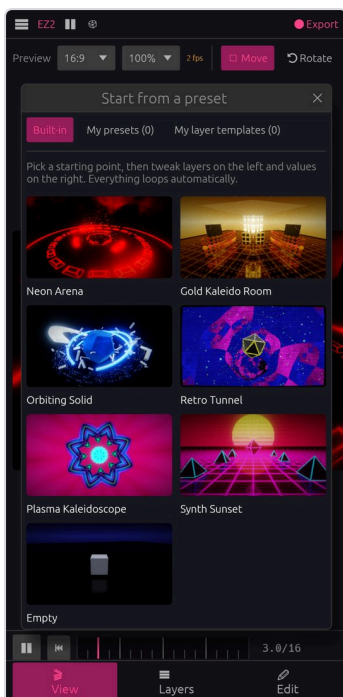
Installing

How	Steps
Browser	Just open the link. The first visit downloads about 5 MB; after that it loads from the cache and also works offline.
Home-screen app (PWA)	In Chrome, open the ⋮ menu → Add to Home screen / Install app . It opens full-screen like a normal app.
Android APK	Download <code>app-debug.apk</code> (or the signed release APK) from the project's GitHub Releases page, open it and allow "install unknown apps" for your browser or file manager when Android asks.

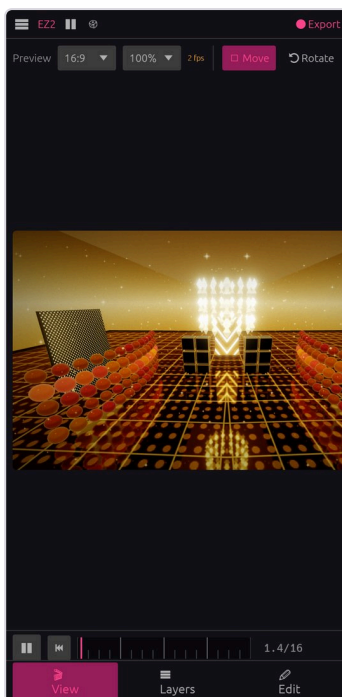
The phone layout

When the window is narrower than about 820 pixels, as on a phone held upright, the editor switches to a **one-panel layout**. The top bar holds **☰** (all menus), play/pause, **🎲** (Surprise me) and **Export**. A compact timeline sits above a **tab bar** with three tabs:

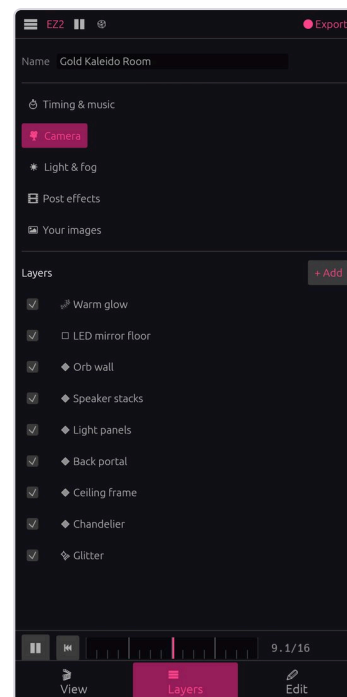
- **View**: the live preview, with the Move/Rotate gizmos.
- **Layers**: the scene settings and layer list (or the **Graph** in Nodes mode).
- **Edit**: the inspector for whatever you selected. Tapping a layer, a scene item or a node jumps here automatically.



Preset gallery on a phone.



The **View** tab.



The **Layers** tab.

Turn a tablet sideways (or use a desktop browser) to get the full desktop layout with all panels side by side.

Touch gestures in the preview

Gesture	What it does
One finger on a gizmo handle	Moves or rotates the selected layer, as with the mouse. Handles are bigger on touch screens.
One finger elsewhere	Orbits the camera around the scene.
Two-finger pinch	Moves the camera closer or further away.
Two-finger twist	Turns the camera around the scene.
Two fingers up/down	Raises or lowers the camera.
Long press on a layer	Opens its menu (duplicate, delete, save as template...), like a right-click.

Files in the browser

- **Importing** models, images and music opens your device's file picker; you can also drop files onto the page on a computer. Imported files are kept inside the browser, so they're still there next time.
- **Saving:** **File** → **Save** downloads a **pack** (`.ez2pack`) with the scene and all its files. Open it again with **File** → **Open...**, on the web or in the desktop app. In the Android app, saving and exporting open Android's **Save as** dialog: pick a folder (Downloads, Documents, Drive...) and tap **Save**. From there you can share the file to a chat with the Files app.
- **My presets, layer templates and the autosave** live in the browser's storage (IndexedDB). If you come back with unsaved changes, the editor offers to **recover** them. Clearing the browser's

site data deletes this library, so save packs of anything you want to keep.

Exporting on the web

Format	Notes
MP4 / WebM video	Encoded by the browser's own video encoder (WebCodecs, hardware-accelerated on most phones). Loops stay frame-exact. MP4 uses H.264 where available, otherwise VP9 or AV1. The web version exports video without the music track ; use the desktop app for videos with sound.
GIF	Works everywhere; loops forever by itself. Keep it small (e.g. 480×270) for sharing.
PNG frames (.zip)	Every frame as a lossless PNG, for editing elsewhere.

The preview pauses while exporting so the device can concentrate on the frames. Phones are slower than desktop PCs, so start with 720p or less, and keep the tab open until the download starts.

WebGPU and WebGL

On computers, recent Chrome and Edge render with **WebGPU**, the fastest path. Other browsers, and all Android phones and tablets, use **WebGL2**, which looks the same (some phone GPUs corrupt the picture with WebGPU). If a scene stutters, lower **Preview** to 50% in the View tab; exports are unaffected.

Graphics settings

☰ → 🖼️ **Graphics** (the 🖼️ button on a computer) shows which backend, GPU and driver the app is using; 📋 **Copy** copies the details for a bug report. On the web and in the Android app you can also force **WebGL2** or **WebGPU** there. The change applies after ↻ **Restart now**. If the chosen backend can't start, the app goes back to **Automatic** by itself.

16 Recipes

Neon arena (black gloss and red neon)

1. Background **Nebula clouds**, with very dark colours A/B and a red C. Set the fog colour to almost-black red.
2. Add a **Mirror floor**: colour black, Reflection 0.35, Blur 0.3.
3. Add **Crystal cluster** with Copies → Radial (count 50, radius 34), Size 7, Random tilt 18, Random size 0.5. Use a glossy black material (metallic 0.9, roughness 0.15, Faceted).
4. Add **Panel** with Radial (20, radius 11) and Stretch 2.6 / 0.35 / 1.2. Glow red, **Edges**, strength 5, and Light chase 1.
5. Add **Ring band** (Width 0.02) stretched 16 / 1 / 16 with red glow, Full, and a Sine animation.
6. Add **Vortex** particles in orange → red. Turn on Bloom 1.1 and Chromatic aberration.

Golden mirrored room

1. Background **Gradient sky** in warm golds. Set the light to warm white and the ambient to 0.45.
2. Add a **Mirror floor** with Reflection 0.85, Reflection tint gold, texture `led_grid` and Grid glow 0.6.
3. Add **Sphere** with Copies → Wall (14 × 5, curve 50°), Size 0.33, Rotation y 90, Position x 6.5, Symmetry **Mirror X**, and Light chase 0.6.
4. Add **Cube** with texture `speaker`, a Grid of 2 × 3 and Symmetry Mirror X. Pulse the glow on the beat.
5. Add **Octahedron** with Spiral copies high above the floor as a chandelier, plus **Snow / glitter** star particles.
6. Post: **Mirror split** left → right and Bloom 1.0.

Crystal orbited by debris

1. Background **Starfield**.
2. Add **Dodecahedron**, Size 1.6, blue, metallic 0.5, roughness 0.18, Faceted, Rim light 0.9, Spin / loop x 1, y 1.
3. Add **Cube** with Copies → Orbit swarm (70, radius 3.2, spread 0.9), Stretch 0.12 / 0.12 / 0.5 and Random size 0.6.
4. Add **Ring band** arcs (Arc 250°) with blue glow and Spin / loop y 2, plus **Ring orbit** particles with trails.

90s demo tunnel

1. Background **Raymarched tunnel** with Wall texture `xor` and Motion / loop 4.
2. Add **Icosahedron** with texture `checker`, Chunky pixels, Glow Edges, and Spin / loop 2 / 1 / 0.
3. Add **Warp stars** with the Pixel square sprite and Trail 4.
4. Post: Pixelate 3, Retro palette EGA with dithering 0.7, CRT monitor, and Beat flash 0.15.

17 Reference

Keyboard & mouse

Action	Shortcut
Play / pause	Space
Save	Ctrl + S
Open	Ctrl + O
Export	Ctrl + E
Undo / Redo	Ctrl + Z / Ctrl + Shift + Z (or Ctrl + Y)
Move / Rotate / Scale tool	W / E / R
Ground grid on/off	G
Select an object	Click it in the viewport
Snap while dragging a handle	Hold Ctrl
Turn the camera	Drag in the viewport (not on a handle)
Zoom the camera	Mouse wheel over the viewport
Scrub time	Click or drag in the timeline
Import	Drop project, model, image or music files onto the window

Performance

The small **fps** number above the viewport shows how smoothly the preview runs. It turns orange below 24 fps. **Hover it** to see the frame time, triangle and particle counts, whether the mirror floor is drawing the scene a second time, and the **five most expensive layers**. A  next to a layer in the list marks a heavy layer. Hover it for details. The usual fixes are fewer copies, less sphere detail, fewer particles or a shorter trail. You can also lower the preview quality, which only affects the preview and never your exports.

Troubleshooting

Problem	Solution
Playback stutters	Hover the fps readout to find the heaviest layers (see Performance above), or lower the preview quality. The fractal background and big particle counts are the heaviest. Exports are not affected.
The computer crashed and I lost my work	Start EZ2DEMOSCENE again and choose Recover (chapter 14).

A friend's project can't find its files	Ask them to send a pack (File → Save as pack) instead of the <code>.ez2.json</code> file.
Everything is washed out or one colour	A big glowing object is probably filling the screen. Reduce Glow, raise the Bloom threshold, or check the Size/Stretch of ring and panel layers.
A shape is invisible	Check its visibility box, its Size, whether it is below the floor or inside another object, and the fog density.
My model shows as a cube	The file couldn't be read. The reason appears in red at the bottom of the inspector. Use Edit → Reload models & images after fixing the file.
No sound	No audio output device was found. The music is still used for sync, ♪ modulation and export.
Video export fails	Check ffmpeg (chapter 13), or export a PNG sequence instead.
The picture corrupts or flickers on a phone or in a browser	Open  Graphics and switch to the other backend (WebGL2 or WebGPU), then restart (chapter 15).
The app doesn't start	Update your graphics drivers. On Linux, install Vulkan drivers (e.g. <code>mesa-vulkan-drivers</code>).

Project files

Projects are saved as readable `.ez2.json` text files. Imported models, images and music are stored as **file paths**, so keep the files inside the project's folder (they're then stored relative to it), or share a **pack** (chapter 14). The built-in presets are also available as project files in `assets/presets`.

Glossary

Term	Meaning
Beat / BPM	A musical pulse. BPM is beats per minute.
Loop / phase	The repeating clip. Phase is a position in it, from 0 (start) to 1 (end, which equals the start).
Instancing	Drawing many copies of one shape efficiently.
Emission / glow	Light a surface gives off by itself.
Bloom	The halo that bright things spill onto their surroundings.
Raymarching	A technique that draws shapes defined by maths (tunnels, fractals) instead of polygons. It is a demoscene classic.
Dithering	Patterns of pixels that fake more colours than a palette has.
Demoscene	A computer art subculture, dating back to the 1980s, that makes real-time audiovisual "demos".